

Innovation Hackathon Preliminary Proposal

Environmental Hacks - Air Track **Team Name:** ICE

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Project Title: AeroRoute - Lower-Exposure Routes Within Your Time Budget

Problem Definition Daily route choices can affect the pollution encountered during a commute. Travel time alone does not capture this exposure, while a longer route through cleaner air may still increase cumulative exposure. Available monitoring data also do not resolve every street. The challenge is to turn imperfect air-quality information into useful route comparisons without presenting unsupported precision.

Target Impact AeroRoute will help pedestrians compare practical commuting options using estimated PM2.5 exposure and journey time. The initial pilot will cover one Delhi area selected after checking monitoring-data availability. The intended benefit is an informed route choice with an explicit time budget; actual exposure reductions will require field validation.

Proposed Solution AeroRoute is a responsive web app for desktop and mobile browsers. A user enters an origin, destination and maximum acceptable detour. The app retrieves walking-route candidates, divides each route into segments and combines estimated PM2.5 concentrations with segment travel times. It presents the fastest route and the candidate with the lowest estimated exposure that meets the time limit.

Estimated Exposure = Sum of (Segment PM2.5 x Segment Travel Time)

Concentration is expressed in micrograms per cubic metre and time in minutes, giving an exposure metric in **microgram-minutes per cubic metre**. This is cumulative ambient exposure, not inhaled dose or an AQI-based health guarantee. The selected route must stay within the fastest journey time plus the user's allowed detour.

Core Contribution The product combines duration-aware route ranking, a detour constraint and visible data quality. Each comparison will show travel time, estimated exposure, source timestamp and coverage limitations. When the available evidence cannot reliably distinguish routes, the app will state that the difference is uncertain.

Hackathon Deliverable One working browser-based flow for walking and PM2.5: enter a journey, compare available routes and inspect the trade-off. Recommendations apply only to evaluated candidates. Any displayed percentage reduction will be labelled as a model estimate; unsupported street-level precision will not be claimed.

Methodology & Tech Stack The prototype will separate the web interface, data processing and route scoring through a compact API.

- **Web Frontend:** Next.js (React), TypeScript and Tailwind CSS with Mapbox GL JS for interactive maps, route overlays and a detour control. A responsive layout will compare time, exposure and data freshness across desktop and mobile browsers. [7]
- **Backend & Database:** Python FastAPI with Supabase PostgreSQL/PostGIS for spatial queries and caching keyed by travel mode, time and data version.
- **Estimation Model:** Start with distance-weighted station interpolation. Evaluate a gradient-boosted regressor using measured station-hour PM2.5 labels, road features, weather and time. A CNN remains a later extension, conditional on sufficient labels and a demonstrated improvement.
- **AWS Deployment:** Host the Next.js frontend on AWS Amplify and the containerized FastAPI service on Amazon Lightsail. Use Amazon S3 for model artifacts and timestamped data snapshots. [1, 5, 8]

Data Sources Access, coverage and freshness will be verified before selecting the pilot area.

- **Ground Measurements:** OpenAQ PM2.5 observations, with station, unit, timestamp and provider metadata. Historical observations supply training labels; current readings anchor estimates. [2]
- **Roads & Routes:** OpenStreetMap/OSMnx for road context; Mapbox Directions for walking geometry and travel times. Mapbox may return up to two alternatives; the system will handle fewer candidates. Walking routes do not directly provide driving-traffic congestion annotations. [3]
- **Weather & Satellite Context:** Open-Meteo weather supplies wind, temperature and humidity. Sentinel-5P NO2 columns and aerosol index are optional regional features, not direct street-level PM2.5 measurements. [4, 6]

Validation & Reliability Compare model error with the interpolation baseline using held-out stations and later time periods; report MAE/RMSE, data coverage and route-response latency. Station-level validation will not establish street-level accuracy. Test route rankings under plausible concentration errors, enforce the detour limit and suppress unsupported claims of improvement. Missing or stale data will trigger a visible limited-data state. Any recorded-data replay will be clearly labelled.

Feasibility & Execution The three-person team will divide ownership across frontend, data/model work and backend/AWS integration. During the four-day build window, the sequence is data access and routing, exposure scoring, integration and validation, then a three-minute recorded demo. The demo will show a complete comparison, the AWS backend and a case where a longer route fails to reduce estimated exposure. [1]

[1] [Environmental Hacks brief](#) [2] [OpenAQ measurements](#) [3] [Mapbox Directions](#)

[4] [Earth Engine: Sentinel-5P NO2](#) [5] [AWS Lightsail containers](#)

[6] [Open-Meteo weather data](#) [7] [Mapbox GL JS](#) [8] [AWS Amplify: Next.js](#)

Proposal scope and outcomes are planned, not yet validated.